REAL-TIME INTRUSION DETECTION SYSTEM

Machine Learning Network Security Report

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Reporting Period: All Recorded Traffic

1. Executive Summary

This report summarizes the network traffic analyzed by the Real-Time Intrusion Detection System. The system uses machine-learning-based classification to identify network flows as either normal traffic or potential intrusions.

Metric	Value
Total Network Flows	12144
Normal Flows	11497
Detected Attacks	647
Attack Rate	5.33%

2. Machine Learning Model

Parameter	Value
Algorithm	Random Forest Classifier
Training Features	duration, protocol_type, src_bytes, dst_bytes, count
Classification	Normal / Attack
Optimized Threshold	0.50
Validation Accuracy	0.76%
Attack Precision	0.96
Attack Recall	0.60
Attack F1-Score	0.74

3. Network Protocol Distribution

Protocol	Flow Count
TCP	3927
UDP	3222
HTTPS	3212
DNS	951
QUIC_OR_UDP443	741
HTTP	50
IPSEC	24
DHCP	13
OTHER	2
ICMP	1
NTP	1

4. Detected Attack History

Time	Source	Destination	Protocol	Threat Vector	Severity	Confidence
2026-09-14 15:07:17	104.18.39.21:443	192.168.1.10:60671	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 15:07:16	140.82.113.21:443	192.168.1.10:65461	HTTPS	ANOMALOUS FLOW	HIGH	94.0%
2026-09-14 15:07:15	192.168.1.10:50694	34.36.133.20:443	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 15:07:11	192.168.1.10:50302	4.186.9.165:443	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 15:07:10	192.168.1.10:50849	142.251.157.4:443	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	78.0%
2026-09-14 15:07:09	192.168.1.10:57272	142.251.154.119:443	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	94.0%
2026-09-14 15:07:02	104.18.39.21:443	192.168.1.10:60671	HTTPS	ANOMALOUS FLOW	HIGH	98.0%
2026-09-14 15:06:47	140.82.113.21:443	192.168.1.10:65461	HTTPS	ANOMALOUS FLOW	HIGH	90.0%
2026-09-14 15:06:43	104.18.39.21:443	192.168.1.10:60671	HTTPS	ANOMALOUS FLOW	HIGH	94.0%
2026-09-14 15:06:42	192.168.1.10:57272	142.251.154.119:443	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	90.0%
2026-09-14 15:06:22	104.18.39.21:443	192.168.1.10:60671	HTTPS	ANOMALOUS FLOW	HIGH	90.0%
2026-09-14 14:36:37	192.168.1.10:54914	142.251.150.4:443	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	85.0%
2026-09-14 14:36:28	192.168.1.10:64561	140.82.114.22:443	HTTPS	ANOMALOUS FLOW	HIGH	92.89%
2026-09-14 14:36:25	192.168.1.10:62302	35.190.80.1:443	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 14:36:24	192.168.1.10:61464	142.251.153.119:443	QUIC_OR_UDP_443	ANOMALOUS FLOW	HIGH	95.0%
2026-09-14 14:36:22	192.168.1.10:63346	172.253.158.188:443	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 14:36:20	192.168.1.10:60772	34.54.84.110:443	HTTPS	ANOMALOUS FLOW	HIGH	100.0%
2026-09-14 14:36:20	192.168.1.10:56794	172.217.119.4:443	HTTPS	ANOMALOUS FLOW	HIGH	76.11%
2026-09-14 14:36:20	3.6.199.65:443	192.168.1.10:54385	HTTPS	ANOMALOUS FLOW	HIGH	95.0%
2026-09-14 14:36:20	3.6.199.65:443	192.168.1.10:55496	HTTPS	ANOMALOUS FLOW	HIGH	95.0%

5. Security Recommendations

1. Investigate repeated attack predictions from the same source address.

2. Monitor network flows with unusually high traffic volume or packet counts.

3. Review suspicious TCP and UDP connections identified by the ML classifier.

4. Use the attack confidence score together with network context when validating alerts.

5. Continue collecting representative network traffic to improve the IDS model.

6. Periodically retrain and evaluate the model using updated datasets.

6. Conclusion

The Real-Time Intrusion Detection System combines live network packet capture, flow-level feature extraction, machine-learning classification, threshold optimization, CSV logging and dashboard-based monitoring. The generated statistics and attack history provide a summary of the traffic observed during the monitoring period.